Introduction to Chemical Engineering and Sustainability (3 credit hours)

Lectures 45 lectures (50 minutes each)

1. The complete chemical engineering curriculum and introduction to the department (1 Lecture)
2. Sustainability in Chemical Engineering - Energy (8 Lecture) -Alan
    a. Renewable Energy
    b. Special Guest Speaker
    c. Hands-on activity
3. Sustainability in Chemical Engineering - Air (8 Lecture) - Vikas
    a. Greenhouse gas emissions: History and Properties
    b. CO2 (1 Lecture)
    c. Acid Rain
    d. Ozone layer protection
    e. Indoor Air
    f. Other Air Pollutants
4. Sustainability in Chemical Engineering - Water (8 Lecture)
    a.
5. Sustainability in Chemical Engineering - Food (8 Lecture) -Betul
    a. Fertilizers
    b.
    c. Coffee - Design
        i. Labs
        ii. Best Coffee Competition
6. Sustainability in Chemical Engineering - Health (8 Lecture) -Betul
    a.
7. Lab Tours
    a. Presentations by faculty
    b. 15 min talks X 2 faculty + lab visit
8. Nanotechnology lab experiment

Credits:

1. What credits will be replaced
   Organic II - 3 credit hours (CHEM 234)
    a. Designed more for pre-med
    b. Too specific for chemical synthesis.
    c. Not relevant for chemical industry - Industrial Mentors (Senior Design)
    d. CHEM 232 includes Structure, reactivity, and synthesis of organic molecules.
        i. Molecular Representation
        ii. Acids & Bases

iii. Alkanes & Cycloalkanes
iv. Stereoisomerism
v. Infrared Spectroscopy
vi. NMR Spectroscopy
vii. Chemical Reactivity & Mechanism
viii. Alkyl Halides (start)
ix. Addition to Alkenes
x. Alkynes
xi. Radical Reactions
xii. Synthesis
xiii. Alcohols & Phenols
xiv. Alcohols & Phenols
xv. Ethers & Epoxides

e. CHEM 233 includes Introductory organic chemistry laboratory. Basic techniques (distillation, crystallization, chromatography, MP and BP), reactions (substitution, elimination, Diels-alder, oxidation-reduction), instrumentation (gas and liquid chromatography, IR.).
f. CHE 234
   i. Mass Spectrometry
   ii. Organometallic Compounds
   iii. Alcohols and Epoxides
   iv. NMR and IR spectroscopy
   v. Diels-Alder & Pericyclic Reactions
   vi. Aromaticity & Electrophilic Aromatic
   vii. Substitution
   viii. Cross Coupling Reactions
   ix. Aldehydes & Ketones
   x. Carboxylic Acids & Derivatives
   xi. Enols & Enolates
   xii. Polymers
   xiii. Radicals
   xiv. Amino acids, Peptides and Proteins
g. Topic that the faculty think should be

Course Title: Introduction to Chemical Engineering and Sustainability
Course Code: CHE 101

Course Description:
This course provides an introduction to the principles and practices of sustainable chemical engineering, with a focus on five key themes: sustainable air, sustainable water, sustainable food, sustainable energy, and sustainable health. Students will explore the challenges and solutions related to sustainability in these areas, applying chemical engineering principles and techniques. The course aims to develop an understanding of the role of chemical engineering in addressing sustainability issues and promoting a sustainable future.

Course Objectives:
1. Understand the concept of sustainability and its relevance to chemical engineering.
2. Gain knowledge of the environmental, social, and economic challenges related to sustainable air, water, food, energy, and health.
3. Develop an understanding of the role of chemical engineering in mitigating environmental impacts and promoting sustainable practices in these areas.
4. Acquire problem-solving skills and critical thinking abilities specific to sustainable chemical engineering.
5. Explore innovative solutions and technologies for sustainable air, water, food, energy, and health.

Course Duration: 16 weeks (1 semester)

Assessment Methods:
1. Quizzes and Examinations: Regular quizzes will be conducted throughout the semester to assess understanding of sustainable engineering concepts. A midterm and a final examination will evaluate overall knowledge and comprehension.
2. Assignments: Problem-solving assignments will be given to apply sustainable engineering principles to real-world scenarios.
3. Projects: Students will work on projects related to sustainable practices in air, water, food, energy, and health.
4. Class Participation: Active participation in class discussions, presentations, and group activities will be considered.

Grading Criteria:
The final grade will be based on a combination of assessments, with weights assigned as follows:
- Quizzes: 20%
- Midterm Examination: 25%

- Assignments: 15%
- Projects: 20%
- Final Examination: 20%

Course Outline:
Each week will consist of lecture content, activity, discussion on movie/Youtube video/ article, Guest speaker, lab tour. Active learning, hand-on experiences and team based learning will be implemented into course.

Week 1: Introduction to Sustainable Chemical Engineering
- Definition and importance of sustainability in chemical engineering
- Overview of the five sustainable themes: air, water, food, energy, and health
- Role of chemical engineering in addressing sustainability challenges
- Activity: What is Chemical Engineering? Who are chemical engineers? (write a story/ drawing)
- Potential Guest Speaker: UOP Renewable Energy

Week 2-3: Sustainable Air (potential faculty: Meenesh)
- Air pollution and its impacts
- Chemical engineering approaches to air quality improvement
- Sustainable technologies for emission reduction
- Case studies on air pollution control in industries
- Activity: Design a air pollution sensor? Or design air filtration system?
- Potential Guest Speaker: EPA

Week 4-5: Sustainable Water (potential faculty: Brian)
- Water scarcity and water pollution challenges
- Water treatment and purification processes
- Membrane technology for sustainable water management
- Case studies on sustainable water use in chemical processes
- Activity: Design a water purification system?
- Potential Guest Speaker: EcoLab/Nalco

Week 6-7: Sustainable Food (potential faculty: Vivek)
- Sustainable agriculture and food production systems
- Food processing and waste reduction strategies
- Packaging and preservation techniques for food sustainability
- Case studies on sustainable food supply chains
- Activity: Coffee Design by UC Davis
- Potential Guest Speaker: Tetra Pak

Week 8-9: Sustainable Energy (potential faculty: Sangil, Said, Ezinne)
- Renewable energy sources and technologies
- Energy efficiency and conservation measures
- Integration of renewable energy in chemical processes
- Case studies on sustainable energy applications in chemical engineering
- Activity: Design a battery, design solar cell (Sanjay's design)
https://academiccollegeprojects.com/renewable-energy-projects-for-engineering-students/
- Potential Guest Speaker: Lanzatech, faculty U of Michigan
(https://che.engin.umich.edu/research/research-areas/sustainable-energy/)

Week 10-12: Sustainable Health (potential faculty: Ying and Vikas)
- Environmental health and chemical exposure
- Green chemistry in pharmaceutical and healthcare industries
- Sustainable waste management in healthcare facilities
- Case studies on sustainable practices for health and well-being
- Activity: https://users.rowan.edu/~hesketh/hesketh/cee%20drug%20delivery.pdf
- Potential Guest Speaker:Abbvie

Week 13-15: Innovative Solutions for Sustainability (potential faculty: Ezinne)
- Emerging technologies for sustainable air, water, food, energy, and health
- Life cycle assessment and sustainability evaluation methods
- Sustainable process design and optimization
- Technological advancements and future prospects
- Activity: Plant Tour
- Potential Guest Speaker: AIChE Sustaianability Institute

Week 16: Review and Final Examination
- Comprehensive review of course topics
- Final examination

Faculty Commitment and Tasks/ Responsibilities:

- One faculty member recommended to teach the course for consistency and engaging students. If many faculty members teach, students will loose engagement and get lost which defeats the purpose of the course.
- Faculty member who is teaching the course is responsible for preparing the content. Other faculty members have advisory role to design the contents in the corresponding module.
- In the delivery of the modules, faculty members recommended to attend at least one lecture time (guest speaker, activity, etc).

- Faculty members are recommended to give a lab tour.
- During the semester, course might have social activities sessions. Faculty members are recommended to attend those activities.
- Overall commitment: 5-7 hours during course design and 2-3 hours during the semester when course is taught.

RESOURCES:

Movies and TV shows:

1. Sustainable Air:
   Movies:
   - "Erin Brockovich" (2000) - Directed by Steven Soderbergh
   - "A Civil Action" (1998) - Directed by Steven Zaillian
   TV Shows:
   - "The Newsroom" (2012-2014) - Created by Aaron Sorkin
   - "Years of Living Dangerously" (2014-2016) - Executive produced by James Cameron

2. Sustainable Water:
   Movies:
   - "Chinatown" (1974) - Directed by Roman Polanski
   - "The Big Blue" (1988) - Directed by Luc Besson
   TV Shows:
   - "Tales by Light" (2015-2018) - Produced by National Geographic
   - "Blue Planet II" (2017) - Narrated by David Attenborough

3. Sustainable Food:
   Movies:
   - "Food, Inc." (2008) - Directed by Robert Kenner
   - "Ratatouille" (2007) - Directed by Brad Bird
   TV Shows:
   - "Chef's Table" (2015-present) - Created by David Gelb
   - "Anthony Bourdain: Parts Unknown" (2013-2018) - Hosted by Anthony Bourdain

4. Sustainable Energy:
   Movies:
   - "Who Killed the Electric Car?" (2006) - Directed by Chris Paine
   - "The 4th Revolution: Energy Autonomy" (2010) - Directed by Carl-A. Fechner
   TV Shows:
   - "Years of Living Dangerously" (2014-2016) - Executive produced by James Cameron
   - "Planet of the Humans" (2019) - Directed by Jeff Gibbs

5. Sustainable Health:
Movies:
- "Contagion" (2011) - Directed by Steven Soderbergh
- "Patch Adams" (1998) - Directed by Tom Shadyac
TV Shows:
- "The Doctors" (2008-present) - Created by Jay McGraw
- "The Mind, Explained" (2019) - Produced by Vox and Netflix

YouTube Videos:
1. Sustainable Air:
- "The Problem with Air Pollution" by TED-Ed
- "Air Pollution 101" by National Geographic
- "How Does Air Pollution Affect Our Health?" by SciShow

2. Sustainable Water:
- "The Water Crisis - Explained" by Kurzgesagt – In a Nutshell
- "How Does Water Pollution Affect You?" by TED-Ed
- "The Story of Bottled Water" by The Story of Stuff Project

3. Sustainable Food:
- "The Story of Food Waste" by National Geographic
- "The Future of Food" by BBC Ideas
- "The Impact of Our Food Choices" by TED-Ed

4. Sustainable Energy:
- "Renewable Energy 101: Solar Power" by National Geographic
- "How Does Wind Energy Work?" by SciShow
- "The Power of Ocean Waves" by TED-Ed

5. Sustainable Health:
- "How Does Climate Change Affect Your Health?" by TED-Ed
- "Sustainable Healthcare: Reducing Waste in Hospitals" by The Green Blue Book
- "Green Hospitals: Contributing to a Sustainable Future" by World Health Organization

Articles
1. Sustainable Air:
- "Air Pollution and Public Health: A Guidance Document for Risk Managers" by World Health Organization

- "Achieving Sustainable Air Quality: An Introductory Guide for Policy Makers" by United Nations Environment Programme
- "Air Pollution Control Technologies for Sustainable Development" by Ajay Kumar Singh and Rakesh Kumar

2. Sustainable Water:
- "Water Scarcity and Sustainable Water Management" by UNESCO
- "Water Treatment Technologies for Sustainable Development" by Saravanamuthu Vigneswaran and Huu Hao Ngo
- "Advances in Water Resources Engineering for Sustainable Development" by K. Srinivasa Raju and R. Jayabalou

3. Sustainable Food:
- "Sustainable Food Systems: A Conceptual Framework" by Jessica Fanzo et al.
- "Sustainable Agriculture and Food Systems: An Integrated Approach" by Olivier De Schutter
- "Food Loss and Waste: Causes, Impacts, and Solutions" by Food and Agriculture Organization of the United Nations

4. Sustainable Energy:
- "Renewable Energy Sources for Sustainable Development" by Ashok V. Mujumdar and N.V. Desai
- "Integration of Renewable Energy Sources: Challenges and Solutions" by Subhes C. Bhattacharyya
- "Sustainable Energy Systems: Pathways for Australian Engineering" by Mark Diesendorf

5. Sustainable Health:
- "Sustainability in Healthcare: An Overview of Sustainable Practices in the Healthcare Sector" by Lawrence D. Prybil
- "Sustainable Healthcare: Concepts, Practices, and Frameworks" by Michael A. Koh et al.
- "Green Chemistry and Sustainable Healthcare: Opportunities and Challenges" by Francesco Ferlin and Emanuela Bozzini

Movies:
1. "An Inconvenient Truth" (2006)
2. "Before the Flood" (2016)
3. "WALL-E" (2008)
4. "Chasing Ice" (2012)
5. "The 11th Hour" (2007)
6. "Food, Inc." (2008)

7. "Into the Wild" (2007)
8. "The Cove" (2009)
9. "Blackfish" (2013)
10. "The Day After Tomorrow" (2004)

TV Shows:
1. "Planet Earth" (2006)
2. "Blue Planet II" (2017)
3. "Years of Living Dangerously" (2014-2016)
4. "Our Planet" (2019)
5. "Cowspiracy: The Sustainability Secret" (2014)
6. "Dirty Jobs" (2005-2012)
7. "Cooked" (2016)
8. "Down to Earth with Zac Efron" (2020)
9. "Rotten" (2018-2019)
10. "The World According to Jeff Goldblum" (2019-present)

Reading list:
1. "Sustainable Air":
   - "Air Pollution Control Engineering" by Noel de Nevers
   - "Fundamentals of Air Pollution" by Daniel Vallero
   - "Air Pollution: Health and Environmental Impacts" by Bhola R. Gurjar, Luisa T. Molina, and C.S.P. Ojha

2. "Sustainable Water":
   - "Water Chemistry" by Mark M. Benjamin
   - "Principles of Water Treatment" by Kerry J. Howe and David W. Hand
   - "Water Resources Engineering" by Larry W. Mays

3. "Sustainable Food":
   - "Food Process Engineering and Technology" by Zeki Berk
   - "Food Engineering Handbook: Food Process Engineering" by Theodoros Varzakas and Constantina Tzia
   - "Sustainable Food Production" by Michael Blakeney, Bruce Paton, and Jonathan Gray
   - “Sustainable food processing and engineering challenges” by Charis Galanakis

4. "Sustainable Energy":
   - "Sustainable Energy: Choosing Among Options" by Jefferson W. Tester, Elisabeth M. Drake, Michael J. Driscoll, Michael W. Golay, and William A. Peters
   - "Introduction to Renewable Energy" by Vaughn C. Nelson

- "Energy and Society: An Introduction" by Harold H. Schobert

5. "Sustainable Health":
- "Green Chemistry: An Introductory Text" by Mike Lancaster
- "Sustainable Development in Chemical Engineering: Innovative Technologies" by Vincenzo Piemonte and Giuseppe Iaquaniello
- "Environmental Health: From Global to Local" by Howard Frumkin, Richard J. Jackson, and Andrew L. Dannenberg

10/13/2023 CHE 101 Committee meeting

- First week, talk about the CHE Curriculum
- Balance between technical aspects and
- Mass and momentum and energy balance
- https://www.amazon.com/Introduction-Chemical-Engineering-Walter-Badger/dp/0070850275 Intro to CHE textbook
- Sustainable chemicals might be another module?
- https://cache.org/teaching-resources-center/introduction-to-chemical-engineering another resource for Intro to CHE
- Comic book https://www.northeastern.edu/landherr/wp-content/uploads/2023/09/ChemEComic-full.pdf
- Guest speakers can be from Alumni, which might be more impactful.
- CHE is the same in terms of unit operations, connecting those unit operations to each module thus linking to curriculum. Traditional vs sustainable
- Foundation of CHE can be introduced in week 1.
- We have to deliver them with the awareness of prerequisite knowledge.
- Write up, lab report, team based? For each module? Projects?
- Budget for activities?
- Having lab component would be great addition.
- Wet lab or classroom? For the course
- https://dl.icdst.org/pdfs/files1/ae44fd68aa54af0e29f1112974fd0522.pdf has a good list of Intro CHE , each topic bring a topic from this list. One by one each
- Name: Intro to CHE with sustainability
- Alan: EPC needs Course title, description, learning outcomes, objectives, ways of achievening course learning outcomes and syllabus.
- Structure of the course: COurse instructor will be in charge of
- Faculty members will be involved in the design of the module activities. All the activity should be in detailed. It might be better to have some standards and structure for the faculty members' visit. It would be better not to talk about research. Faculty members should be asked to attend the activity.

- Having a routine would be helpful: One lecture time, one movie/discussion, one guest speaker and one activity.

1 **Chemical Engineering** 1

2 **Basic Concepts of Analysis** 23

3 **The Balance Equation** 60

4 **Component Mass Balances** 66

5 **Membrane Separation** 81

6 **Chemically Reacting Systems** 96

7 **Designing Reactors** 115

8 **Bioreactors and Nonlinear Systems** 130

9 **Overcoming Equilibrium** 140

10 **Two-Phase Systems and Interfacial Mass Transfer** 144

11 **Equilibrium Staged Processes** 168

12 **Energy Balances** 187

13 **Heat Exchange** 202

14 **Energy Balances for Multicomponent Systems** 217

15 **Energy Balances for Reacting Systems** 233